“A SIMPLE BANKING SYSTEM”

A Project Report submitted
In partial fulfillment of the requirements for the award of the degree
of
BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE & ENGINEERING
(DATA SCIENCE)

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DECLARATION

I hereby declare that the project report entitled “SIMPLE BANKING SYSTEM” is an original work done in the Department ofComputer Science and Engineering (Data Science), GITAM School of Technology, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering. The work has not been submitted to any other college or University for the award of any degree or diploma.

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INTRODUCTION

A Simple Banking System is a software application designed to manage basic banking operations such as depositing money, withdrawing money, and checking account balance. In today's digital world, banks use computerized systems to handle customer transactions efficiently and securely.

This project focuses on developing a simple banking system that helps users perform basic financial activities in an organized manner. Instead of maintaining records manually, the system stores and processes all transaction details digitally. This reduces errors and saves time.

The system allows users to create an account, deposit money, withdraw money, and view their balance whenever required. It provides a simple and user-friendly interface so that even beginners can use it easily.

Overall, the Simple Banking System helps in understanding how banking operations work and demonstrates the use of programming in real-life financial applications.

AIM OF PROJECT

The main aim of this project is to develop a simple and efficient banking system that allows users to perform basic banking operations easily.

The project is designed to help users manage their bank accounts in a systematic and organized way. It allows users to deposit and withdraw money, as well as check their account balance at any time.

Another aim of this project is to provide a user-friendly platform where all banking activities can be performed quickly without confusion. The system simplifies financial transactions and reduces the need for manual record keeping.

This project also aims to demonstrate how programming concepts can be applied to build real-world applications like banking systems.

Overall, the aim is to make banking operations simple, fast, and accurate using a digital system.

PROBLEM DEFINITION AND OBJECTIVES

Problem Definition

In traditional banking, managing account details manually can be time-consuming and prone to errors. Customers may face difficulties in keeping track of their transactions, such as deposits and withdrawals.

Manual systems also make it difficult to maintain accurate records and can lead to mistakes in calculations. In addition, accessing account information quickly becomes challenging without a proper system.

Therefore, there is a need for a simple digital banking system that can store, manage, and process user transactions efficiently. A Simple Banking System solves this problem by providing an easy way to handle banking operations digitally.

Objectives

The main objective of this project is to create a system that allows users to manage their bank accounts effectively.

One objective is to provide basic banking features such as account creation, deposit, withdrawal, and balance checking.

Another objective is to ensure that all transactions are recorded properly so that users can track their financial activities easily.

The system also aims to reduce manual errors by performing automatic calculations for balance updates.

It helps users save time and improves the accuracy of financial data management.

Finally, the project demonstrates the practical use of programming in building simple banking applications.

CODE

```
class BankAccount:

    def __init__(self, name):
        self.name = name
        self.balance = 0
        self.transactions = []

    def deposit(self, amount):
        if amount > 0:
            self.balance += amount
            self.transactions.append(f"Deposited  : +Rs.{amount}")
            print(f"\n  Rs.{amount} deposited successfully.")
        else:
            print("\n  Invalid amount.")

    def withdraw(self, amount):
        if amount <= 0:
            print("\n  Invalid amount.")
        elif amount > self.balance:
            print("\n  Insufficient balance.")
        else:
            self.balance -= amount
            self.transactions.append(f"Withdrawn  : -Rs.{amount}")
            print(f"\n  Rs.{amount} withdrawn successfully.")

    def check_balance(self):
        print(f"\n  Current Balance: Rs.{self.balance}")

    def transaction_history(self):
        print(f"\n  Transaction History for {self.name}:")
        if len(self.transactions) == 0:
            print("  No transactions yet.")
        else:
            for i, t in enumerate(self.transactions, 1):
                print(f"  {i}. {t}")
        print(f"  ----------------------------")
        print(f"  Balance: Rs.{self.balance}")
```

```
def main():
    print("=" * 40)
    print("       WELCOME TO BANK")
    print("=" * 40)

    name = input("Enter your name: ")
    account = BankAccount(name)

    print(f"\n  Hello, {account.name}! Account created.")

    while True:
        print("\n--- MENU ---")
        print("1. Deposit Money")
        print("2. Withdraw Money")
        print("3. Check Balance")
        print("4. Transaction History")
        print("5. Exit")

        choice = input("\nEnter choice (1-5): ")

        if choice == "1":
            try:
                amount = float(input("Amount to deposit: Rs."))
                account.deposit(amount)
            except:
                print("  Invalid input. Enter a number.")

        elif choice == "2":
            try:
                amount = float(input("Amount to withdraw: Rs."))
                account.withdraw(amount)
            except:
                print("  Invalid input. Enter a number.")

        elif choice == "3":
            account.check_balance()

        elif choice == "4":
            account.transaction_history()
```

```
    elif choice == "5":
        print(f"\n  Goodbye, {account.name}!")
        break

    else:
        print("\n  Invalid choice. Try 1-5.")


main()
```

1st option and Output

```
Output
========================================
       WELCOME TO BANK
========================================
Enter your name: Vidya

  Hello, Vidya! Account created.

--- MENU ---
1. Deposit Money
2. Withdraw Money
3. Check Balance
4. Transaction History
5. Exit

Enter choice (1-5): 1
Amount to deposit: Rs.300000

  Rs.300000.0 deposited successfully.

--- MENU ---
1. Deposit Money
2. Withdraw Money
3. Check Balance
4. Transaction History
5. Exit
```

2nd option and Output

```
Enter choice (1-5): 2
Amount to withdraw: Rs.3000

  Rs.3000.0 withdrawn successfully.

--- MENU ---
1. Deposit Money
2. Withdraw Money
3. Check Balance
4. Transaction History
5. Exit

Enter choice (1-5): 3

  Current Balance: Rs.297000.0

--- MENU ---
```

3rd option and Output

```
Enter choice (1-5): 3

  Current Balance: Rs.297000.0

--- MENU ---
1. Deposit Money
2. Withdraw Money
3. Check Balance
4. Transaction History
5. Exit
```

4th option and Output

```
Enter choice (1-5): 4

  Transaction History for Vidya:
  1. Deposited  : +Rs.300000.0
  2. Withdrawn  : -Rs.3000.0
  ----------------------------
  Balance: Rs.297000.0

--- MENU ---
1. Deposit Money
2. Withdraw Money
3. Check Balance
4. Transaction History
5. Exit
```

5th option and output

```
--- MENU ---
1. Deposit Money
2. Withdraw Money
3. Check Balance
4. Transaction History
5. Exit

Enter choice (1-5): 5

  Goodbye, Vidya!

=== Code Execution Successful ===
```

Future Work

The Simple Banking System can be improved by adding more advanced features in the future to make it more useful and secure.

One major improvement is the addition of multiple user accounts so that different users can access their own accounts with unique login credentials. This will make the system more realistic and suitable for real-world use.

Another enhancement is the introduction of a graphical user interface (GUI). A GUI will make the system more interactive and easier to use compared to a text-based interface.

The system can also include transaction history, where users can view all past deposits and withdrawals. This will help users track their financial activities more clearly.

Security features such as password protection and authentication can be added to protect user data from unauthorized access.

In the future, the system can also support online banking features like fund transfer between accounts. This will make the system more advanced and closer to real banking systems.

Integration with databases can also be implemented so that data is stored permanently and can be accessed anytime.

The system can further be developed into a mobile application, allowing users to perform banking operations on their smartphones.

Overall, these improvements will make the Simple Banking System more efficient, secure, and user-friendly

Conclusion

The Simple Banking System is an effective solution for managing basic banking operations in a digital way. It allows users to deposit and withdraw money, check their balance, and maintain their account records easily.

This system reduces the complexity of manual banking and provides a faster and more accurate way of handling financial transactions. It also helps users understand how banking systems work in real life.

The project demonstrates the importance of technology in improving financial services and making them more accessible.

Overall, the Simple Banking System is a useful application that simplifies banking operations and can be further enhanced with advanced features in the future.